Supplementary Material for "SensitiveAI: An Ablation Study of Optimization Recipes for Vietnamese Hate Speech Detection"

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Abstract. This supplement is the technical appendix to the main paper. It fixes the protocol, lists the fixed data split, and reports analyses that were summarised only in one or two sentences in the main text: the complete per-class precision/recall/F1 matrix and raw confusion counts, one-vs-rest ranking quality (ROC-AUC and average precision) and the Matthews correlation coefficient, posterior calibration, an error-complementarity analysis over all 2^6 wrong/right patterns of the six systems, a stratified macro-F1 by surface phenomenon, the complete 15-pair significance matrix, a post-hoc decision-rule diagnostic, and the resolved hyperparameters and wall-clock cost. Every number below is computed from the frozen, per-run test predictions; all tables and figures are regenerated by a single script (`scripts/11_supplementary_analyses.py`). This document is a supplement and is not part of the main paper page budget.

Keywords: Hate speech detection · reproducible ablation · error analysis · calibration.

1 Overview and Scope

The main paper compares six systems (M1–M6) that share one backbone (`microsoft/mdeberta-v3-base` [4,1]), one fixed three-way split of the ViHSD corpus [3], one label mapping (HATE / OFFENSIVE / CLEAN) and one random seed, and differ only in the optimisation recipe. All six systems were trained once and scored once on the held-out test set of 6,680 comments. This supplement does not train or tune anything new: it re-reads the frozen per-run artefacts (`test_predictions.csv` with the three class posteriors, `metrics.json`, `training_summary.json`) and derives additional diagnostics from them. Because the posteriors are frozen, none of the diagnostics here can influence the headline numbers of the main paper; they are descriptive analyses of *how* the six recipes differ, not a search for a better recipe.

The main text reports a pre-specified subset of seven pairwise comparisons and a small set of summary metrics. This appendix reports the *complete* matrices

and reconciles them with the main text where the larger view adds information (in particular, the complete accuracy significance matrix in Sect. 10 contains more surviving pairs than the seven-pair subset, and it is stated explicitly which is which).

2 Data and the Fixed Split

Every system reads the same split file, whose SHA-256 digest is copied into each run directory and re-checked before training, so no run can silently start from a different split. Table 1 restates the split and its digests; these are the values behind Table 1 of the main paper.

Table 1. The fixed split used by all six systems. Digests are the first 12 hex digits of the full SHA-256 recorded in the released `split_manifest.json`.

Split	Comments	HATE	OFF.	CLEAN	SHA-256
train	24,046	2,556	1,605	19,885	`f705ffe94d41`
validation	2,672	270	212	2,190	`6aa552aafb92`
test	6,680	688	444	5,548	`d8f9ae7ddd8c`

The test set is imbalanced (83% CLEAN), which drives most of the behaviour discussed below: the minority classes HATE and OFFENSIVE are the ones where recipes visibly differ, and they dominate the macro-F1 and the error analyses.

3 Complete Per-Class Results

Table 2 gives the full per-class precision, recall and F1 for all six systems, plus the macro-F1 that the main paper ranks on. The pattern reported in the main text is visible in the individual columns: the focal-only [2] continuation (M6) has by far the best HATE recall (0.6468) and consequently the best HATE F1 (0.6088), but this is paid for by the *lowest* CLEAN recall of the six (0.9216); the multi-view head (M3) has the best OFFENSIVE F1 (0.4561) and the best macro-F1 (0.6638) with a more balanced profile.

Table 2. Per-class precision, recall and F1 on the ViHSD test set for all six systems, with the macro-F1. Best per column in bold.

System	HATE			OFFENSIVE			CLEAN			M-F1
	P	R	F1	P	R	F1	P	R	F1	
M1	0.6446	0.5378	0.5864	0.4343	0.4392	0.4367	0.9314	0.9497	0.9405	0.6545
M2	0.6427	0.5334	0.5830	0.4844	0.4189	0.4493	0.9258	0.9553	0.9403	0.6575
M3	0.6176	0.5727	0.5943	0.4745	0.4392	**0.4561**	0.9339	0.9479	0.9409	**0.6638**
M4	0.6037	0.5669	0.5847	0.4822	0.4279	0.4535	0.9312	0.9466	0.9389	0.6590
M5	0.6417	0.5363	0.5843	0.5075	0.3806	0.4350	0.9233	0.9605	**0.9415**	0.6536
M6	0.5749	0.6468	**0.6088**	0.4098	0.4707	0.4382	0.9476	0.9216	0.9344	0.6604

Table 3 reports the raw (unnormalised) confusion counts, with one row per system and column blocks grouped by *true* class. Reading the OFFENSIVE true block, every system sends a large share to CLEAN; the M6 row is the one that moves mass *out* of CLEAN and into the two minority columns, which is exactly the accuracy/macro-F1 trade-off. The normalised confusion matrices for all six systems are shown together in Fig. 1.

Table 3. Raw test-set confusion counts for all six systems over the 6,680 test comments. The three column blocks are the *true* class and the sub-columns the *predicted* class (H = HATE, O = OFFENSIVE, C = CLEAN); each row therefore sums to 6,680.

	true HATE			true OFF.			true CLEAN		
System	H	O	C	H	O	C	H	O	C
M1	370	98	220	81	195	168	123	156	5,269
M2	367	83	238	71	186	187	133	115	5,300
M3	394	86	208	85	195	164	159	130	5,259
M4	390	84	214	80	190	174	176	120	5,252
M5	369	64	255	87	169	188	119	100	5,329
M6	445	92	151	103	209	132	226	209	5,113

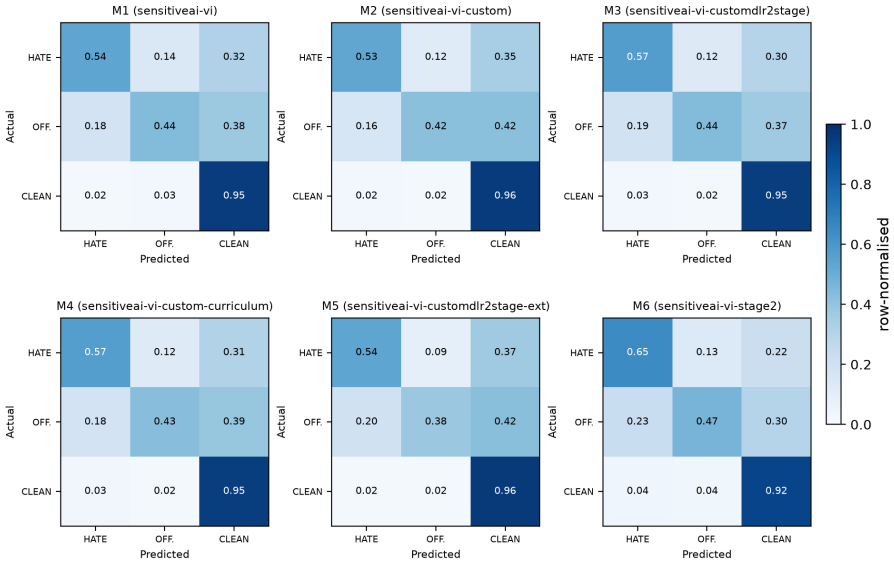


Fig. 1. Normalised confusion matrices (rows = true, columns = predicted) for the six systems, on the same colour scale. M6 (bottom right) trades CLEAN recall for minority recall relative to the other five.

4 Ranking Quality: ROC-AUC, Average Precision and MCC

Accuracy and macro-F1 depend on a decision rule; the one-vs-rest ranking metrics do not, and so they separate "the model ranks the classes well" from "the threshold happened to be good." Table 4 reports per-class ROC-AUC and average precision (AP) [5], their macro and micro averages, and the Matthews correlation coefficient (MCC). The headline reading is that the systems are close but not identical: M6 has the best macro ROC-AUC (0.8930) while M1 has the best macro AP (0.6599), so the best *ranking of the rare positives* and the best *recall of the rare positives* do not belong to the same recipe. MCC, which is insensitive to the CLEAN majority, is highest for M3 (0.5650), agreeing with its best macro-F1.

Table 4. One-vs-rest ranking quality per class (ROC-AUC / average precision) and the macro averages, for all six systems. Best macro value in bold.

	HATE		OFFENSIVE		CLEAN	
System	AUC	AP	AUC	AP	AUC	AP
M1	0.8946	0.5982	0.8686	0.4119	0.8987	0.9698
M2	0.8966	0.5717	0.8635	0.4052	0.8924	0.9684
M3	0.8538	0.5354	0.8532	0.3898	0.8713	0.9511
M4	0.8799	0.5735	0.8002	0.3844	0.8700	0.9522
M5	0.9100	0.5707	0.7221	0.3625	0.7967	0.9042
M6	0.8985	0.5688	0.8724	0.4182	0.9079	0.9731

System	macro AUC / AP	micro AUC / AP	MCC
M1	0.8873 / **0.6599**	0.9635 / 0.9361	0.5541
M2	0.8842 / 0.6484	0.9604 / 0.9332	0.5533
M3	0.8594 / 0.6254	0.9577 / 0.9189	0.5650
M4	0.8500 / 0.6367	0.9528 / 0.9158	0.5550
M5	0.8096 / 0.6124	0.9346 / 0.8481	0.5538
M6	**0.8930** / 0.6534	0.9575 / 0.9311	0.5636

The per-class ROC and precision–recall curves for all six systems are reproduced in Sect. 8; this table is their numerical companion.

5 Posterior Calibration

A system can be accurate yet over-confident, which matters when its scores are used as moderation signals. Table 5 therefore reports top-label expected calibration error (ECE, 10 equal-width bins), the multiclass Brier score and the negative log-likelihood (NLL) of the frozen posteriors, all computed on the test set. M6 is the best-calibrated system on every metric (ECE 0.068, Brier 0.217, NLL 0.482), with a mean confidence of 0.931; the class-weighted cross-entropy systems (M3, M5) are the most over-confident (mean confidence 0.996) and have

Table 5. Calibration of the frozen posteriors on the test set: top-label expected calibration error (ECE, 10 bins), multiclass Brier score and negative log-likelihood (lower is better). Best per column in bold.

System	ECE ↓	Brier ↓	NLL ↓	mean conf.
M1	0.106	0.228	0.622	0.979
M2	0.111	0.233	0.838	0.987
M3	0.120	0.244	1.191	0.996
M4	0.119	0.241	1.044	0.993
M5	0.118	0.237	1.047	0.996
M6	**0.068**	**0.217**	**0.482**	0.931

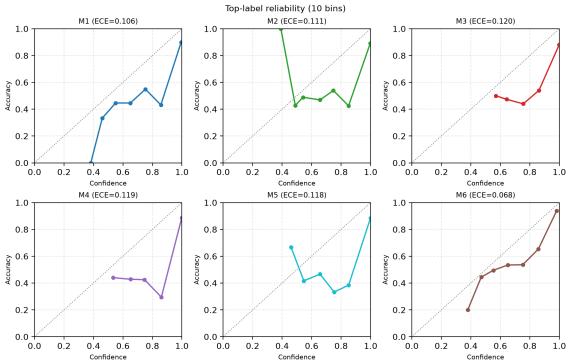


Fig. 2. Reliability diagram of the top-label confidence for the six systems (10 bins). The dashed diagonal is perfect calibration. M6 sits closest to the diagonal; M3 and M5 are the most over-confident.

the worst NLL (> 1.0), i.e. they are frequently very confident and wrong. The reliability diagram is shown in Fig. 2.

6 Error Complementarity

Do the systems make the same mistakes, or different ones? For each test comment we count how many of the six systems misclassify it. Table 6 (top) gives the distribution over $k = 0, \ldots, 6$: 5,410 comments are solved by all six and 504 (7.5% of the test set) are missed by all six, with the remaining 766 split across the middle bins. The 504 all-miss comments are the hard core (HATE 185, OFFENSIVE 180, CLEAN 139) and set a floor that no recipe in this family reaches past. Fig. 3 visualises the same distribution.

Table 6. Error complementarity across the six systems. *Top:* how many test comments are misclassified by exactly k systems. *Bottom:* the comments no system gets right, broken down by their gold label.

# systems wrong	0	1	2	3	4	5	6
test comments	5,410	228	178	93	143	124	504

missed by all six: 504 comments (HATE 185, OFFENSIVE 180, CLEAN 139)
solved by all six: 5410 comments (HATE 270, OFFENSIVE 130, CLEAN 5010)

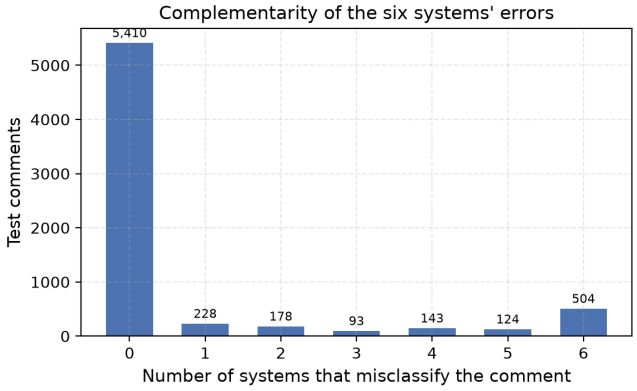


Fig. 3. Number of test comments misclassified by exactly k of the six systems. The spike at $k = 6$ is the 504-comment hard core; the rest is spread thinly, i.e. the systems mostly agree on which items are easy.

Table 7 isolates the opposite quantity: errors made by *exactly one* system. M6 has 109 unique errors, an order of magnitude more than M3 (11) or M2 (25); its errors are both the most numerous (913) and the least shared with the other systems, which is the error-profile counterpart of its minority-recall-heavy operating point.

Table 7. Errors made by exactly one system (out of the six). A high count means the system's mistakes are largely its own; a low count means its errors are shared with the others.

System	unique errors	total errors
M1	43	846
M2	25	827
M3	11	832
M4	20	848
M5	20	813
M6	109	913

7 Stratified Macro-F1 by Surface Phenomenon

Vietnamese social-media text is noisy in ways that are not captured by a single accuracy number. Table 8 recomputes macro-F1 within overlapping surface strata (emoji, teencode, repeated characters, missing/with diacritics, and very short / very long comments) defined by simple lexical rules on the raw comment. Strata overlap, so a comment can appear in several rows. Two patterns stand out: (i) *very long* comments are the hardest stratum for every system (macro-F1 0.52–0.61, $n = 145$) and the largest gap between systems, where M3 leads (0.6118) and M1 lags (0.5212); (ii) on short comments, which are the majority ($n = 1{,}852$ for *very short*), the six systems fall within 0.015 of one another (0.698–0.713). These are descriptive within-stratum gaps on a single test split; we do not attribute them to a causal effect of any surface phenomenon.

Table 8. Macro-F1 by surface phenomenon of the test comments, for all six systems. Strata overlap (a comment can be both short and teencode); counts are the number of test comments carrying the tag.

Stratum	M1	M2	M3	M4	M5	M6	n
`emoji`	0.5826	0.6165	0.6417	0.6397	0.6266	0.6522	974
`no_diacritics`	0.6120	0.6336	0.6382	0.6191	0.6283	0.5882	370
`repeat_chars`	0.6820	0.6988	0.6913	0.6861	0.6783	0.6651	997
`teencode`	0.6527	0.6301	0.6520	0.6480	0.6476	0.6591	1,039
`very_long`	0.5212	0.5677	0.6118	0.6029	0.5256	0.5411	145
`very_short`	0.7098	0.7090	0.7132	0.6981	0.7087	0.7083	1,852
`with_diacritics`	0.6537	0.6562	0.6628	0.6592	0.6531	0.6627	6,114

8 Comparison Figures Across the Six Systems

The figures below are the full set produced by the evaluation pipeline. Each plot compares all six systems on the same axis so that the differences in accuracy, macro-F1, per-class F1, validation loss and validation accuracy, and the ranking quality, can be read off directly. They expand on the summary tables above; nothing here is tuned on the test set. Two readings are worth flagging. First, the accuracy leader is M5 (0.878), not M6 (0.863, the lowest); the gap

reverses on macro-F1, where M5 is last (0.654) and M3 first (0.664), which is the accuracy/macro-F1 trap made visible. Second, the validation-loss curves are *not* the model selection criterion: the selected checkpoint is the epoch with the highest validation macro-F1 (`src/callbacks/monitoring.py`), so the frozen epoch need not be the loss minimum.

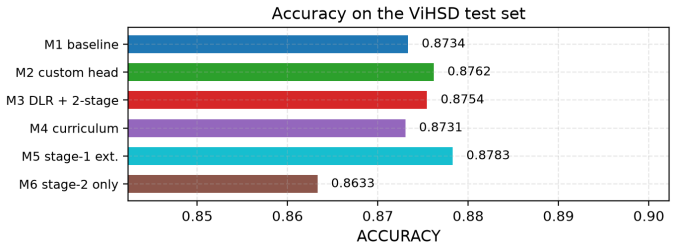


Fig. 4. Test accuracy for all six systems with 95% bootstrap confidence intervals. The highest accuracy belongs to M5, which does not have the best macro-F1.

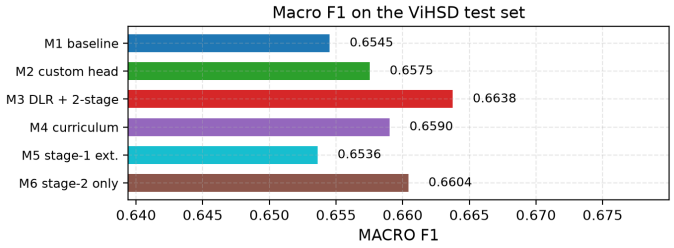


Fig. 5. Test macro-F1 (mean of the three class F1 scores) for all six systems. The six systems sit in a narrow band, which is why no pairwise macro-F1 difference is significant.

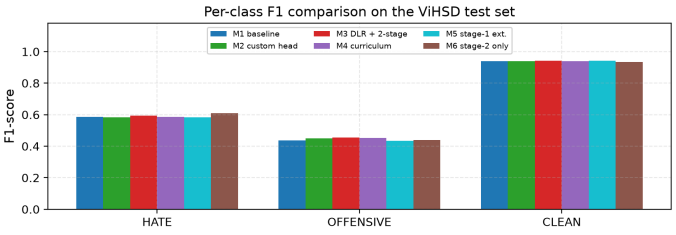


Fig. 6. Per-class F1 for HATE, OFFENSIVE and CLEAN, grouped by system. The minority classes (HATE, OFFENSIVE) show the largest spread across systems, while CLEAN is saturated.

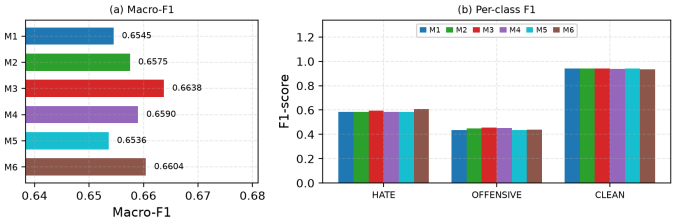


Fig. 7. Macro-F1, weighted-F1 and per-class F1 side by side for every system.

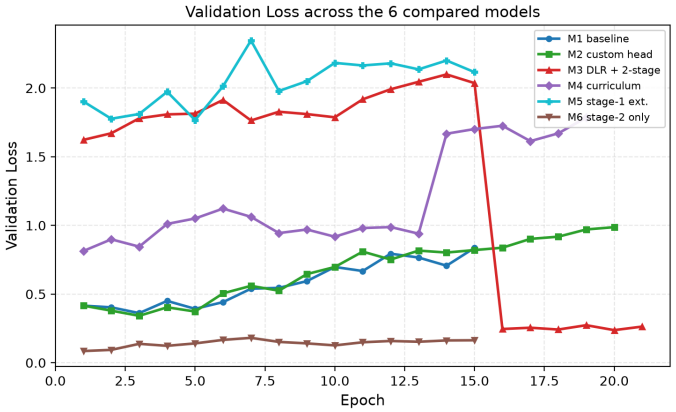


Fig. 8. Validation loss over the training epochs for all six systems. The checkpoint is *not* selected to minimise this curve: it is the epoch with the highest validation macro-F1 (Fig. effig:supp-val-acc), which is why the selected checkpoints can sit on a rising-loss epoch.

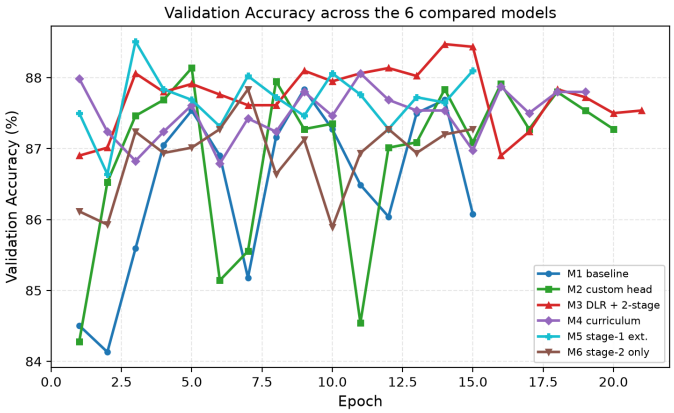


Fig. 9. Validation accuracy over the training epochs for all six systems.

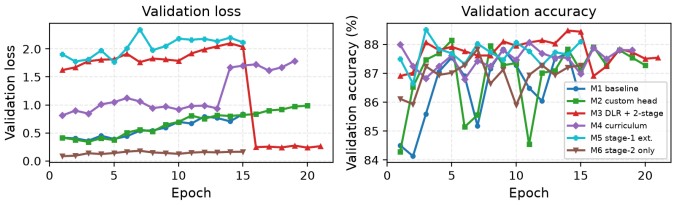


Fig. 10. Validation loss and accuracy on the same axes for all six systems.

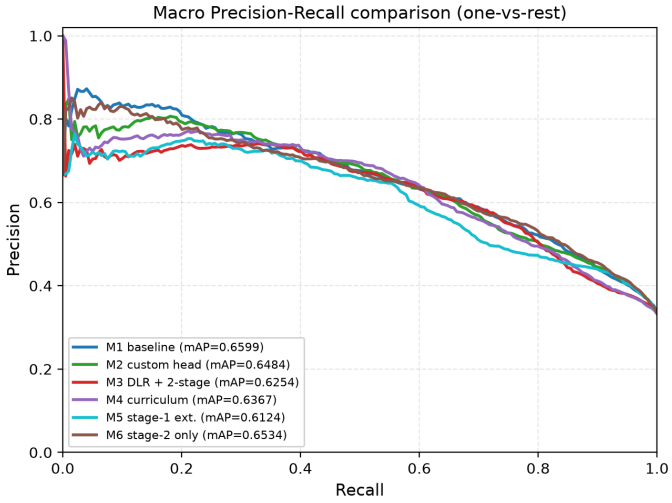


Fig. 11. One-vs-rest precision–recall curves (micro and macro averages) for all six systems.

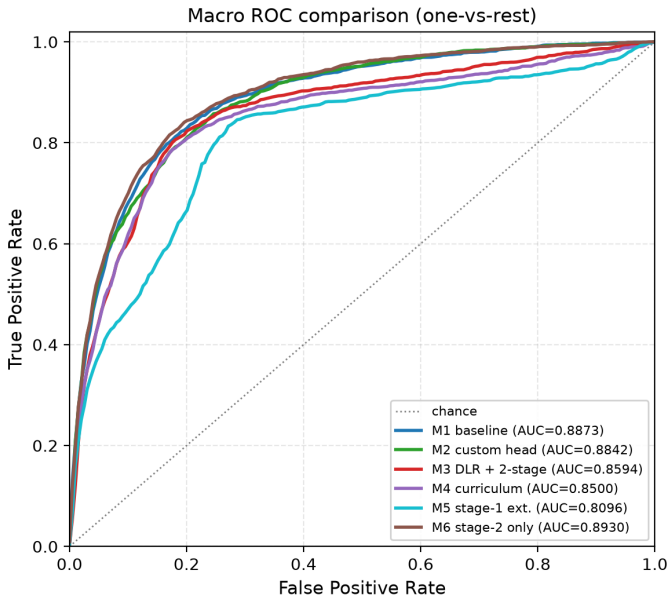


Fig. 12. One-vs-rest ROC curves (micro and macro averages) for all six systems.

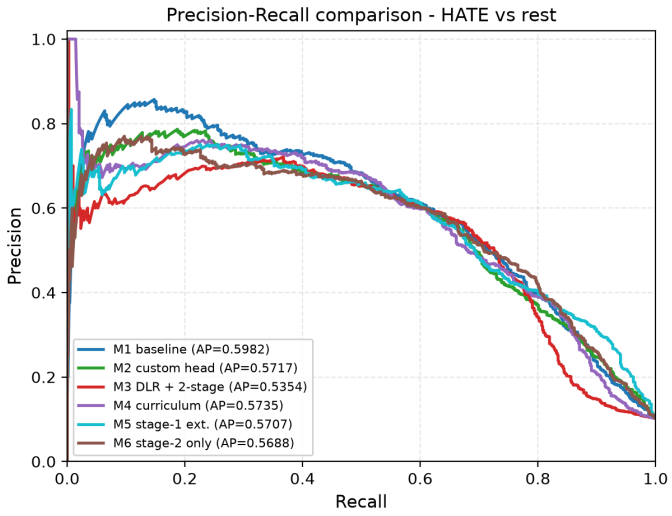


Fig. 13. Precision–recall curve for the HATE class, all six systems.

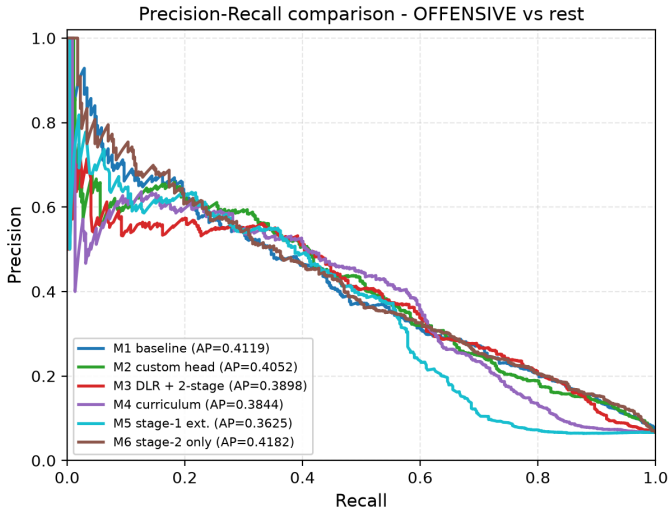


Fig. 14. Precision–recall curve for the OFFENSIVE class, all six systems.

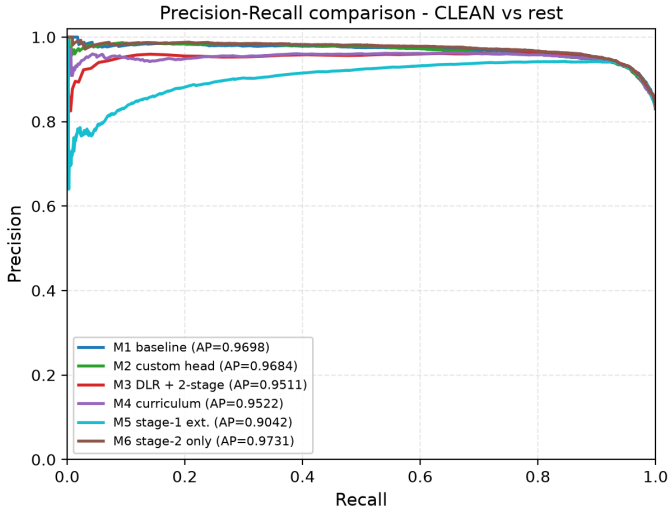


Fig. 15. Precision–recall curve for the CLEAN class, all six systems.

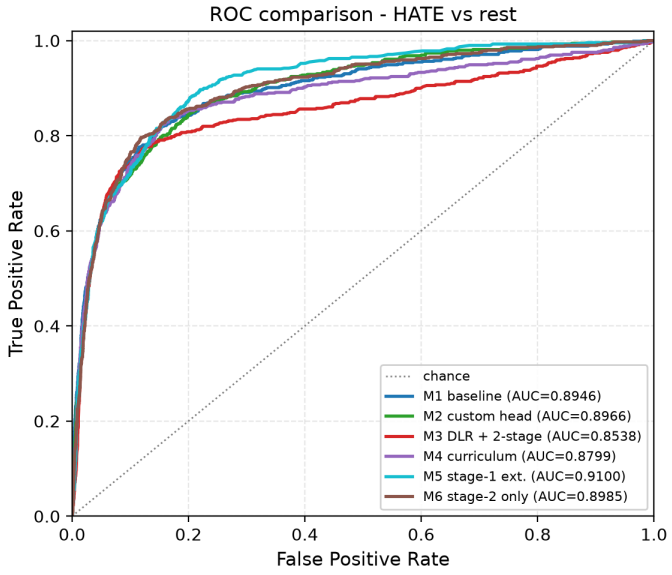


Fig. 16. ROC curve for the HATE class, all six systems.

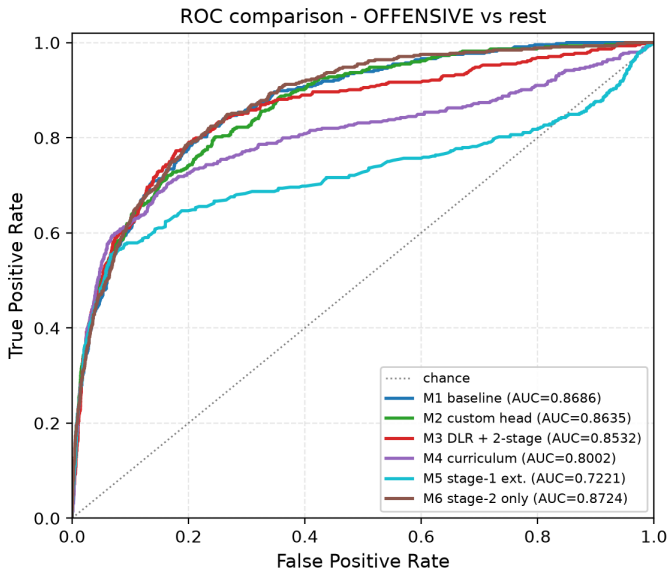


Fig. 17. ROC curve for the OFFENSIVE class, all six systems.

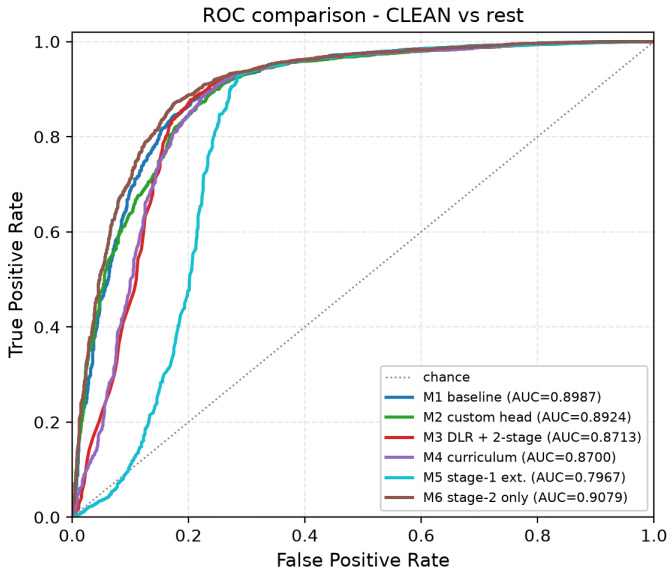


Fig. 18. ROC curve for the CLEAN class, all six systems.

9 Per-System Figures

For each system we reproduce its normalised confusion matrix, its precision–recall and ROC curves, and its validation loss and accuracy curves. These are the raw diagnostic plots for every run, laid out one system at a time so that a reviewer can inspect an individual recipe in isolation.

9.1 M1

Baseline: the pre-trained encoder with a standard [CLS] softmax head, single-stage training and plain cross-entropy. It has the second-best macro average precision (0.6599).

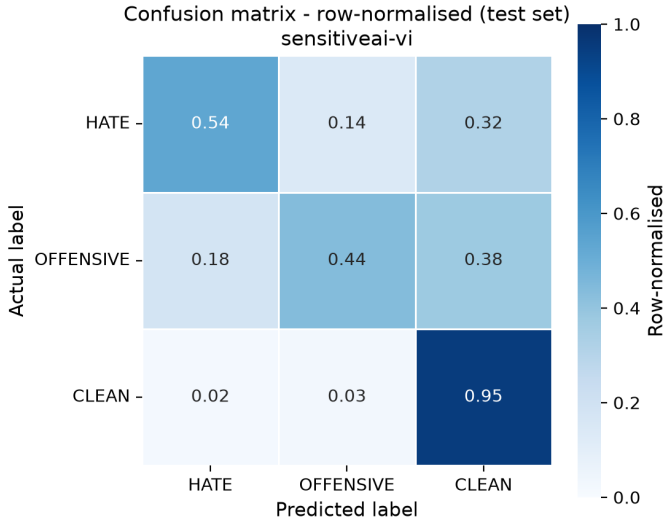


Fig. 19. Normalised confusion matrix (rows = true, columns = predicted).

9.2 M2

Baseline plus a *multi-view pooling* head: the first token (`[CLS]`), masked-mean and masked-max views are concatenated to a 2,304-d vector and passed through a LayerNorm → Linear(2304 → 512) → GELU → dropout bottleneck MLP (512 → 3). There is no convolutional layer and the optional attention-pooling view is disabled; training is single-stage with plain cross-entropy.

9.3 M3

Multi-view pooling head with discriminative learning rates and a second focal-loss stage; the best macro-F1 of the six (0.6638) and the highest MCC. Its threshold search returned the trivial 0.5.

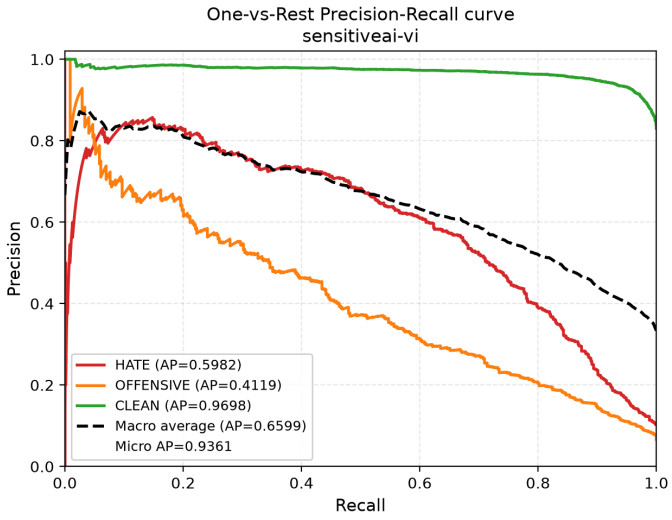


Fig. 20. Precision–recall curve.

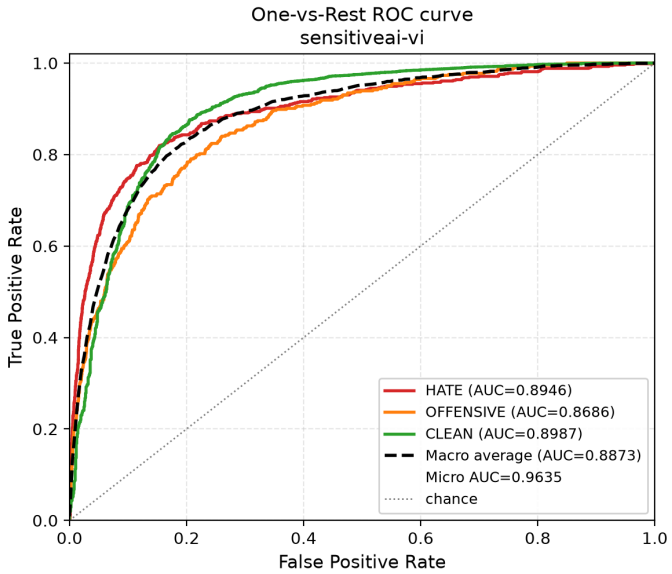


Fig. 21. ROC curve.

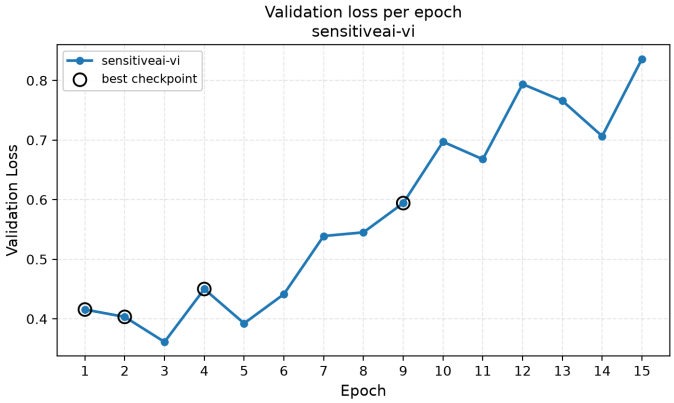


Fig. 22. Validation loss over the training epochs.

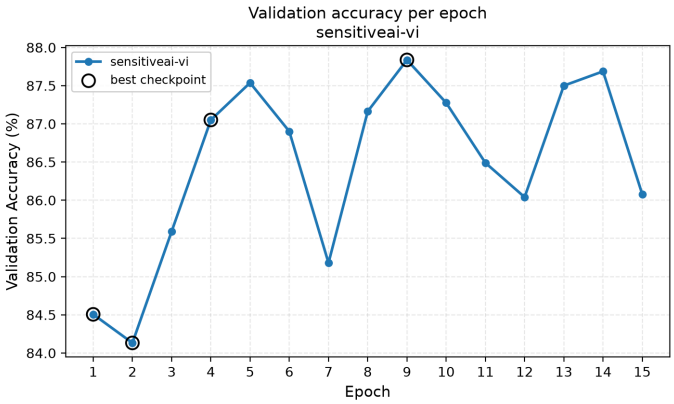


Fig. 23. Validation accuracy over the training epochs.

9.4 M4

Multi-view head trained as a three-stage curriculum, initialised from M3's checkpoint. Stage 1 (concept loss, OFFENSIVE) was allocated 10 epochs but early stopping truncated it after 7; stage 2 (concept loss, HATE) ran 6 (absolute epochs 8–13) and stage 3 (cross-entropy, class weights 3:2:1) ran 6 (absolute epochs 14–19), for 19 of 30 planned epochs. The run's best *validation* macro-F1 (0.6942) occurred at absolute epoch 1 and was never beaten, so the later stages were not selected; its searched thresholds (0.01/0.5/0.64) raised validation macro-F1 but did not transfer to test.

9.5 M5

Continuation of M3's stage-1 recipe (DLR plus class-weighted cross-entropy) for a further 15 epochs, reaching the full budget because the early-stopping patience

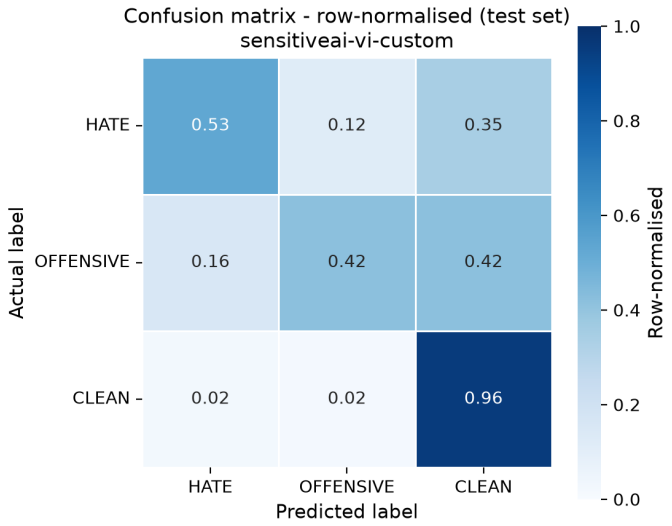


Fig. 24. Normalised confusion matrix (rows = true, columns = predicted).

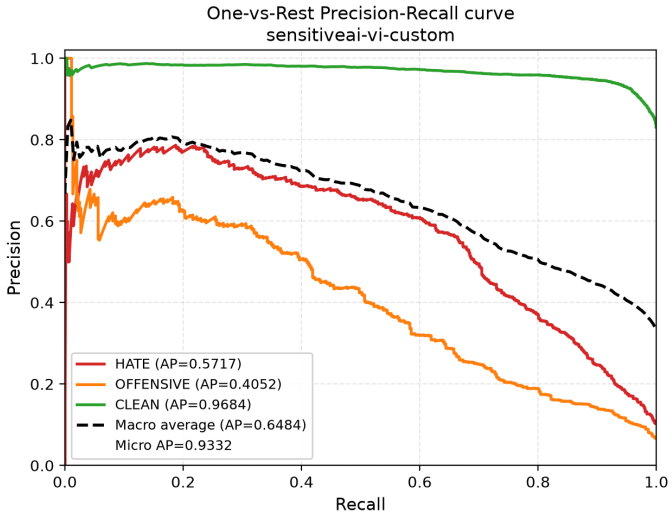


Fig. 25. Precision–recall curve.

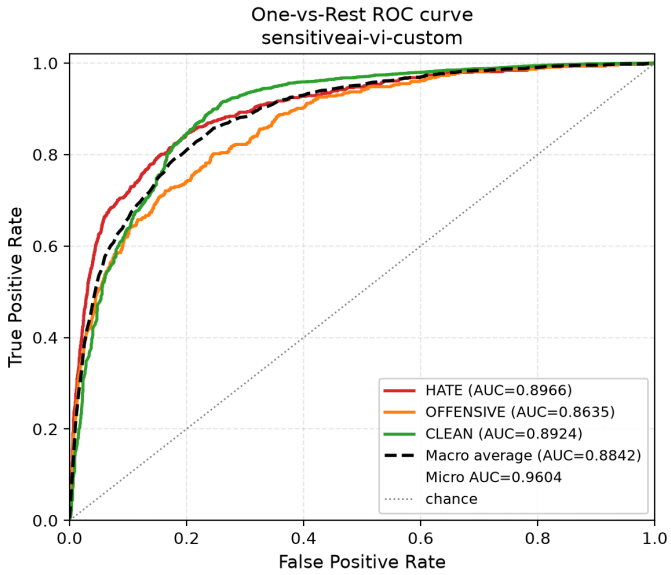


Fig. 26. ROC curve.

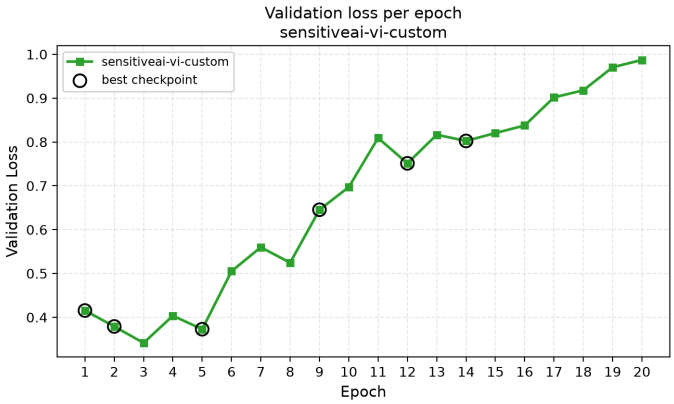


Fig. 27. Validation loss over the training epochs.

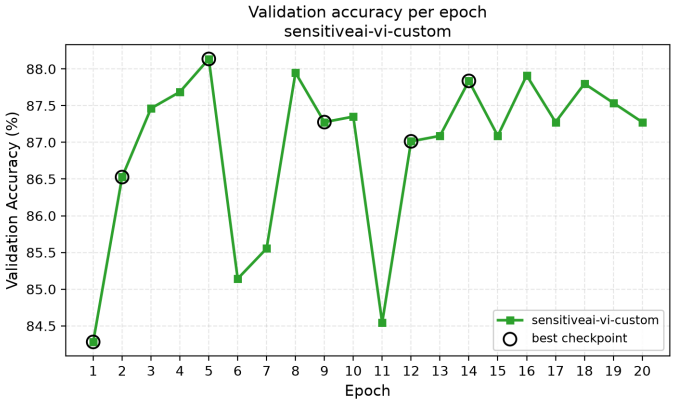


Fig. 28. Validation accuracy over the training epochs.

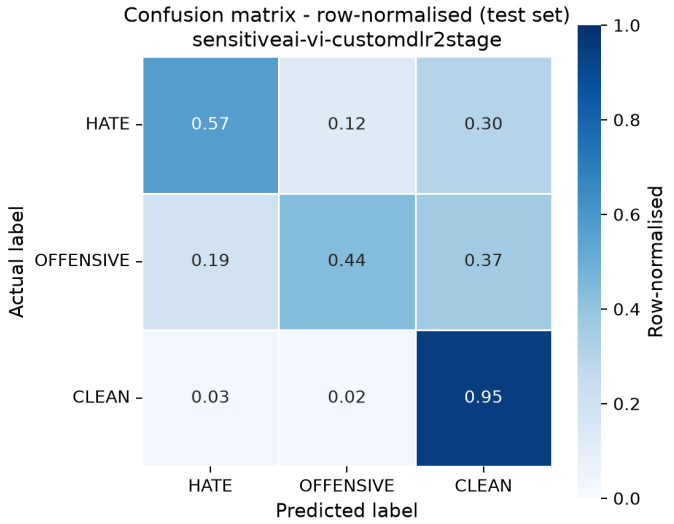


Fig. 29. Normalised confusion matrix (rows = true, columns = predicted).

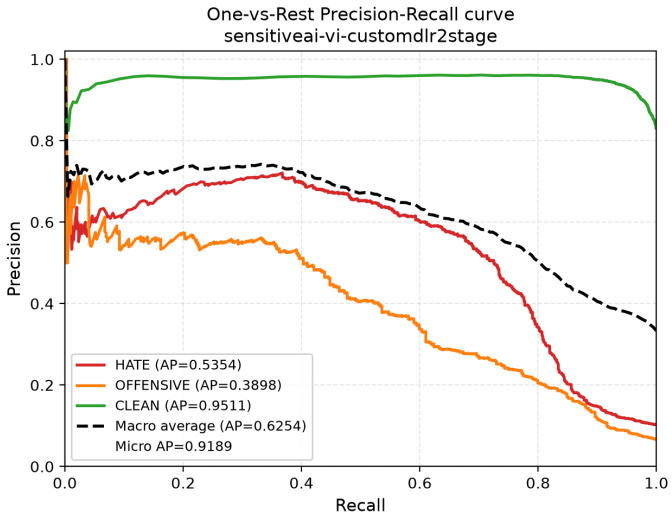


Fig. 30. Precision–recall curve.

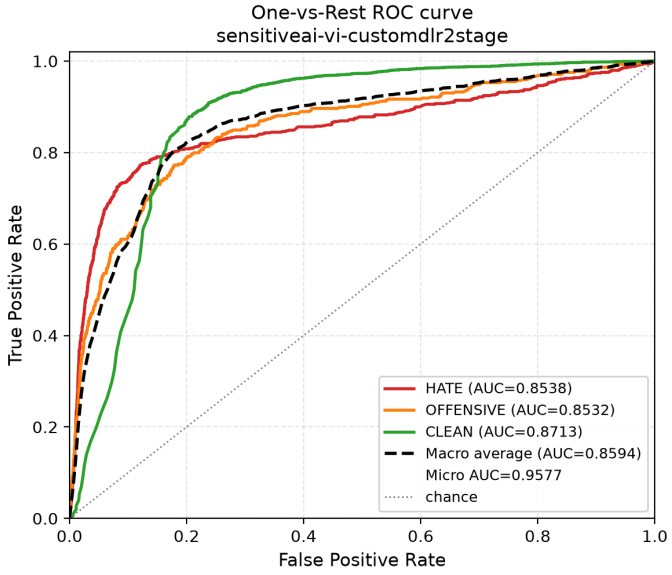


Fig. 31. ROC curve.

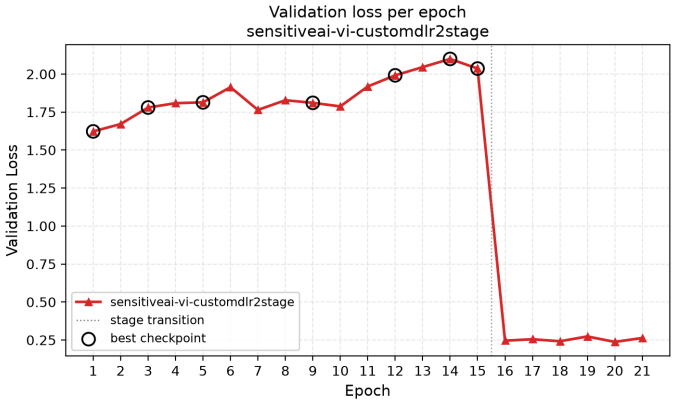


Fig. 32. Validation loss over the training epochs.

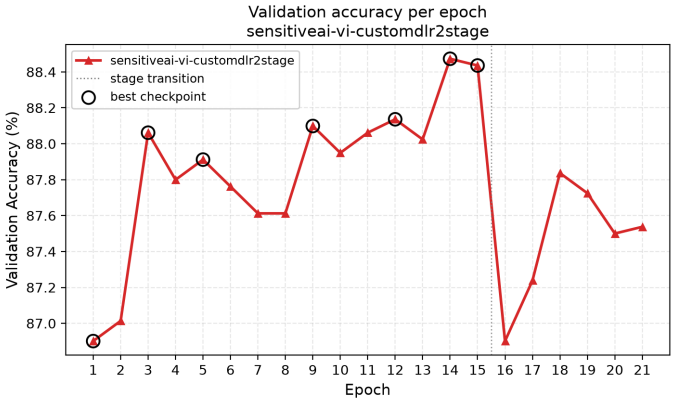


Fig. 33. Validation accuracy over the training epochs.

equals the budget and could not trigger earlier. It has the *highest* test accuracy (0.878) but the *lowest* macro-F1 (0.654)—the accuracy/macro-F1 trap in one system.

9.6 M6

A focal-only second stage on the baseline head (class-weighted focal loss, no multi-view head). It has the best HATE recall (0.647), the best HATE F1 (0.609) and the best macro ROC-AUC (0.893), but the lowest CLEAN recall (0.922) and the lowest accuracy (0.863).

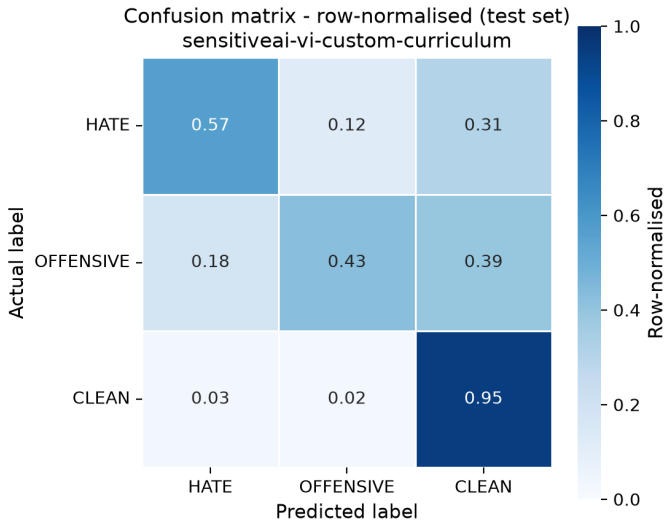


Fig. 34. Normalised confusion matrix (rows = true, columns = predicted).

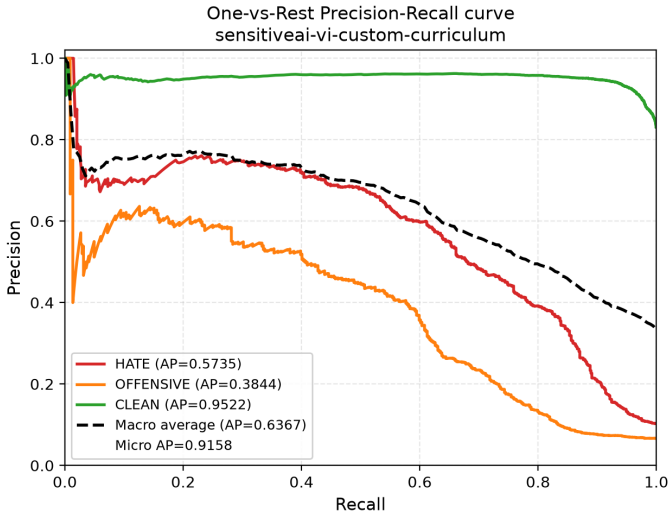


Fig. 35. Precision–recall curve.

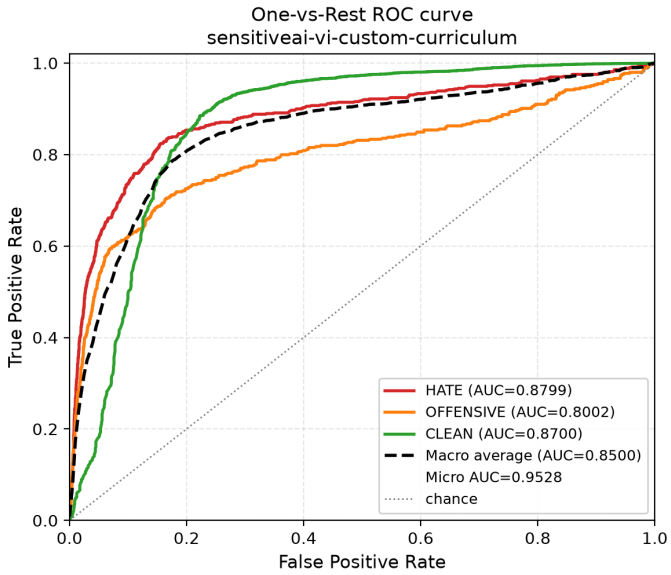


Fig. 36. ROC curve.

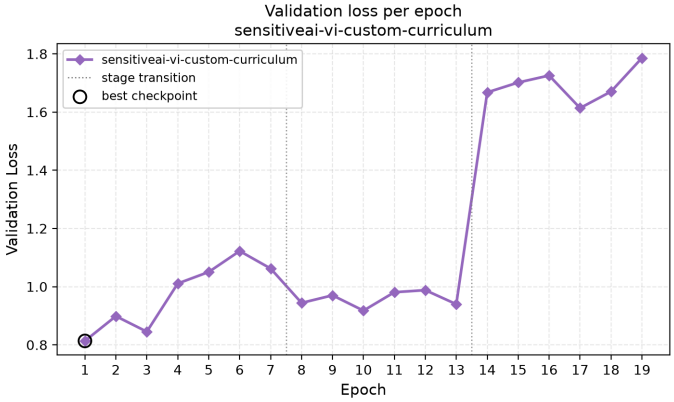


Fig. 37. Validation loss over the training epochs.

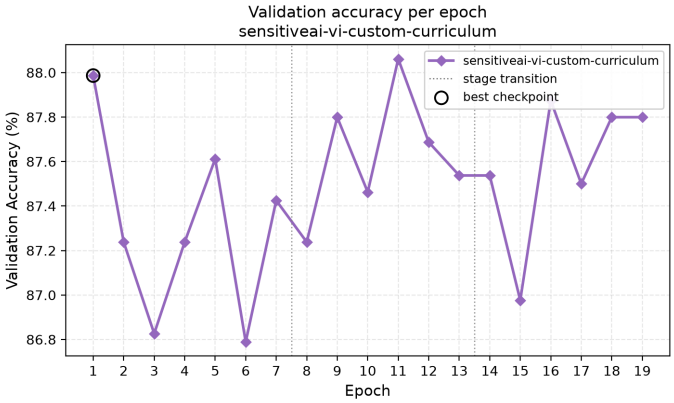


Fig. 38. Validation accuracy over the training epochs.

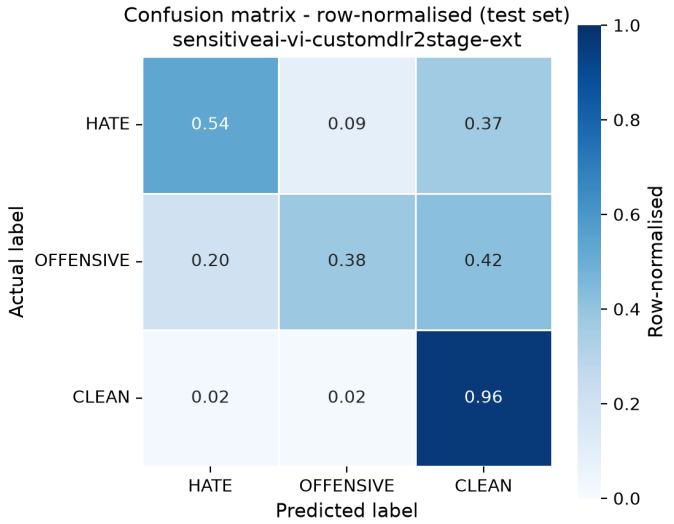


Fig. 39. Normalised confusion matrix (rows = true, columns = predicted).

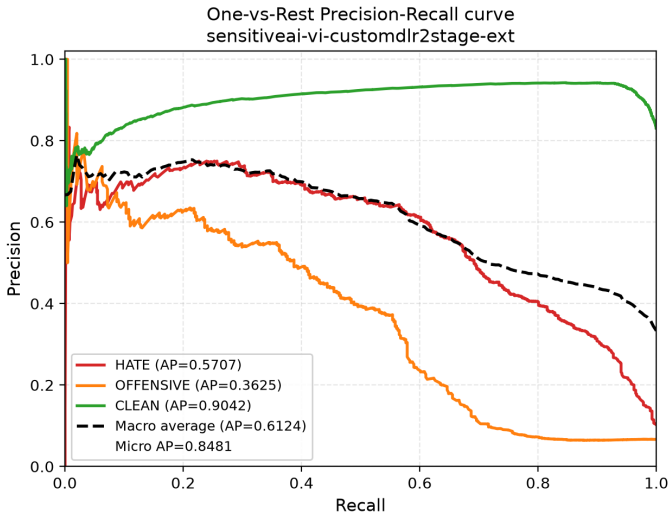


Fig. 40. Precision–recall curve.

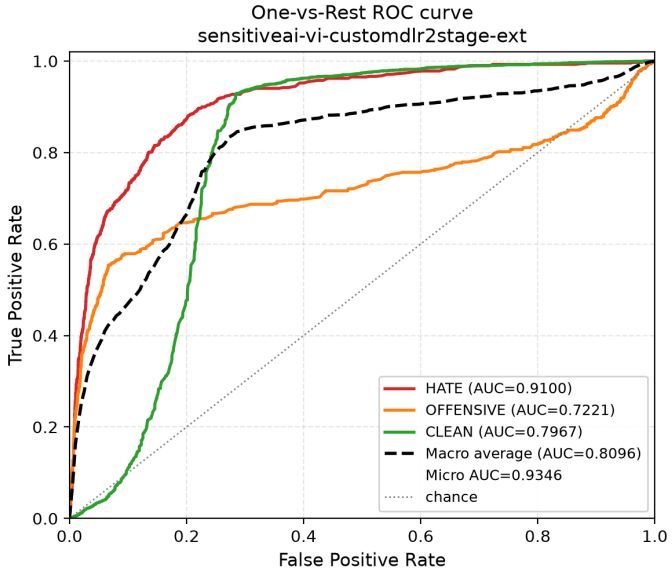


Fig. 41. ROC curve.

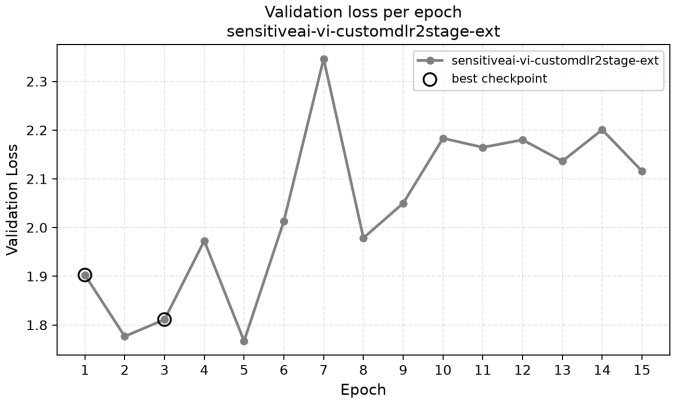


Fig. 42. Validation loss over the training epochs.

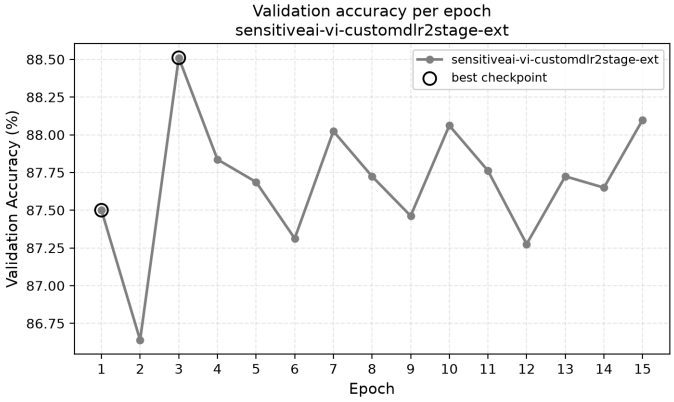


Fig. 43. Validation accuracy over the training epochs.

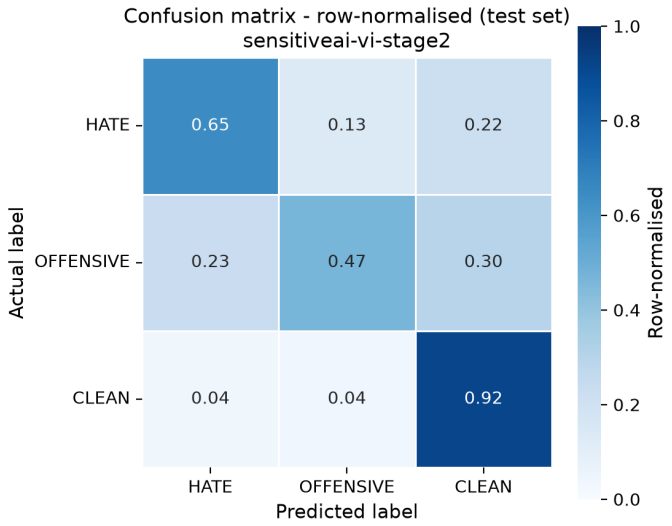


Fig. 44. Normalised confusion matrix (rows = true, columns = predicted).

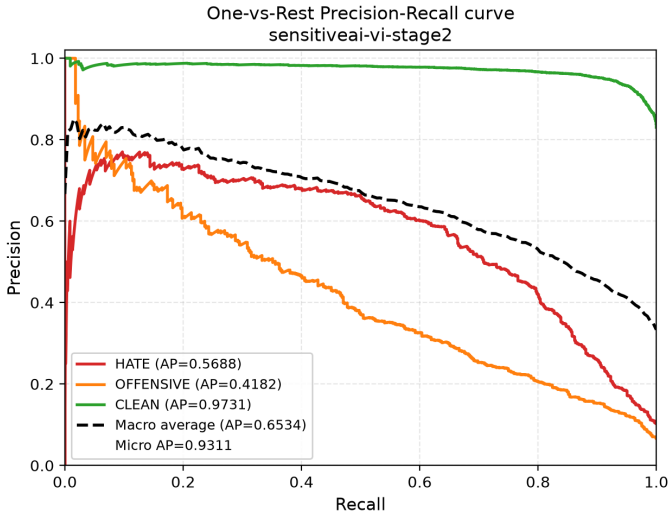


Fig. 45. Precision–recall curve.

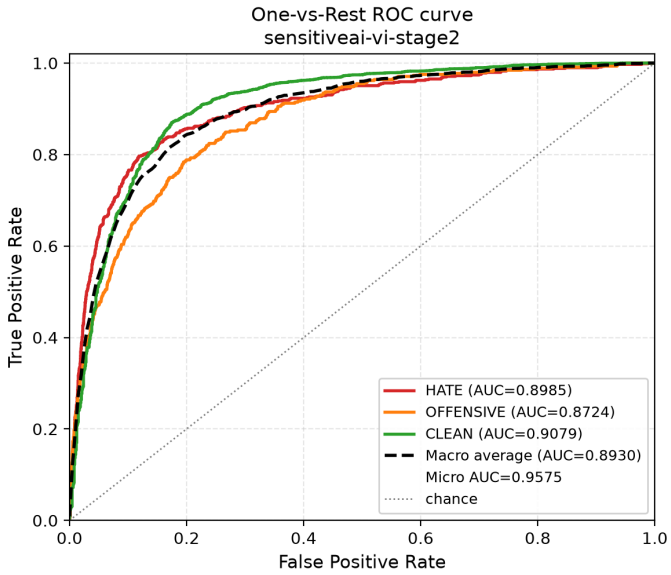


Fig. 46. ROC curve.

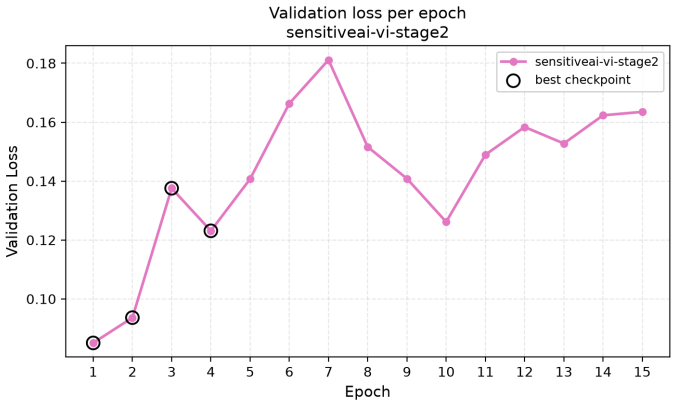


Fig. 47. Validation loss over the training epochs.

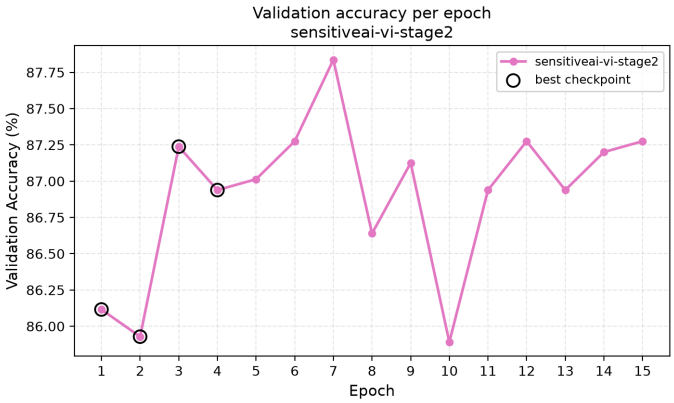


Fig. 48. Validation accuracy over the training epochs.

10 Complete Pairwise Significance Matrix

The main paper reports a pre-specified subset of seven pairwise comparisons. Table 9 gives the *complete* matrix over all $\binom{6}{2} = 15$ pairs: the paired-bootstrap macro-F1 difference with a 95% confidence interval and its p-value, and the exact two-sided McNemar p-value on accuracy, with a Holm–Bonferroni step-down correction over these 15 tests. This complete view confirms and sharpens the main-text claim:

- **Macro-F1.** No pair is significant: every 95% bootstrap interval for Δmacro-F1 contains zero, consistent with the main-text statement that the six systems are statistically indistinguishable on macro-F1.
- **Accuracy.** Every pair that involves M6 is significant after Holm correction ($p_{\text{Holm}} \leq 0.030$), because M6 trades CLEAN recall for minority recall. The main text reported this for the two M6 pairs inside its seven-pair subset (M3–M6 and M6–M1); the complete matrix shows the same effect holds for all five M6 pairs. No pair *not* involving M6 survives Holm in accuracy.

Table 9. Complete pairwise comparison of all $\binom{6}{2} = 15$ system pairs. ΔM-F1 is the paired-bootstrap difference (A minus B) with a 95% CI, p_{F1} its bootstrap p-value; p_{acc} is the exact two-sided McNemar p-value on accuracy. † marks $p < 0.05$ after a Holm–Bonferroni step-down correction over these 15 tests.

$A - B$	ΔM-F1 [95% CI]	p_{F1}	p_{acc} (McN.)	p_{acc} (Holm)
M1 − M2	-0.0030 [-0.0203, +0.0144]	0.736	0.370	1.000
M1 − M3	-0.0092 [-0.0269, +0.0082]	0.298	0.526	1.000
M1 − M4	-0.0045 [-0.0215, +0.0127]	0.594	0.961	1.000
M1 − M5	+0.0009 [-0.0164, +0.0185]	0.906	0.110	0.989
M1 − M6	-0.0059 [-0.0191, +0.0071]	0.377	$< 0.001^{\dagger}$	0.003
M2 − M3	-0.0062 [-0.0186, +0.0060]	0.328	0.788	1.000
M2 − M4	-0.0015 [-0.0137, +0.0107]	0.807	0.186	1.000
M2 − M5	+0.0039 [-0.0093, +0.0174]	0.568	0.385	1.000
M2 − M6	-0.0029 [-0.0196, +0.0135]	0.719	$< 0.001^{\dagger}$	0.002
M3 − M4	+0.0047 [-0.0051, +0.0144]	0.342	0.217	1.000
M3 − M5	+0.0101 [-0.0023, +0.0228]	0.113	0.217	1.000
M3 − M6	+0.0033 [-0.0124, +0.0188]	0.678	$< 0.001^{\dagger}$	0.002
M4 − M5	+0.0054 [-0.0080, +0.0190]	0.421	0.027	0.270
M4 − M6	-0.0014 [-0.0172, +0.0139]	0.859	$0.003^{\dagger}$	0.030
M5 − M6	-0.0068 [-0.0237, +0.0098]	0.419	$< 0.001^{\dagger}$	< 0.001

11 Post-hoc Decision-Rule Diagnostic

Because the main paper finds that a validation-searched threshold rule did not help, it is worth asking how much a decision rule could help in the best case. Table 10 reweights the two minority posteriors by a joint factor u and re-argmaxes, and reports the *oracle* $u^{\star}$ that maximises test macro-F1. Within this

one-parameter reweighting family the oracle gains are tiny (at most +0.0051 for M6, +0.0018 for M1, and exactly 0 for M3 at $u^\star = 1$), well inside the bootstrap confidence intervals of Table 9. The diagnostic therefore bounds what *this* family of decision rules can contribute, and shows that it is not where the remaining error lies; it does not rule out richer, class- or instance-dependent rules, and the ceiling it exposes is a property of the frozen posteriors rather than a claim about the encoder alone. Fig. 49 sweeps the full range of u.

Because $u^\star$ is chosen on the test set, the "best M-F1" column is an upper bound and not a validated decision rule; it is reported only to bound how much thresholding *could* contribute.

Table 10. Post-hoc prior-reweighting diagnostic. The frozen posteriors are reweighted by a joint factor u on the two minority classes and re-argmaxed; $u^\star$ is the *oracle* factor that maximises test macro-F1. "argmax M-F1" is the macro-F1 of a plain argmax on the frozen posteriors; it differs from the system's headline macro-F1 only for M4, whose validation-searched thresholds (0.01/0.5/0.64) *lowered* test macro-F1 from the argmax 0.6602 to the reported 0.6590. $u^\star$ is chosen on the test labels and the gain is therefore an oracle upper bound over this one-parameter reweighting family, not a deployable decision rule.

System	argmax M-F1	$u^\star$	best M-F1	oracle gain
M1	0.6545	0.70	0.6564	+0.0018
M2	0.6575	0.80	0.6585	+0.0009
M3	0.6638	1.00	0.6638	+0.0000
M4	0.6602	2.80	0.6616	+0.0014
M5	0.6536	3.00	0.6549	+0.0012
M6	0.6604	0.60	0.6656	+0.0051

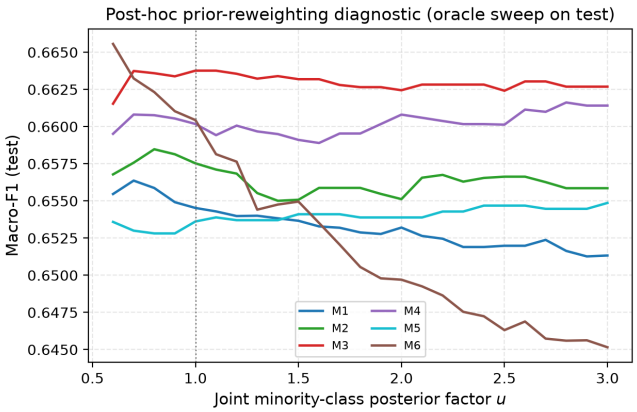


Fig. 49. Test macro-F1 as a function of the minority reweighting factor u for the six systems. The curves are flat near their maxima: no threshold change recovers a meaningful amount of macro-F1.

12 Resolved Hyperparameters and Training Cost

Table 11 records the resolved configuration of each system as actually logged by the run (not as intended), and Table 12 the wall-clock cost and checkpoint selection. The "thr." column is read from each evaluated run's `metrics.json`, which is authoritative for the test pass even where the config on disk was edited later: M1 and M2 applied a plain argmax, M3–M6 applied per-class thresholds searched on validation, and only M4 found non-trivial values (0.01/0.5/0.64). Three reproducibility points are worth stating explicitly.

First, checkpoint selection is by *highest validation macro-F1* (`src/callbacks/monitoring.py`), not by lowest validation loss. M4 selected its *first* epoch (validation macro-F1 0.6942) and never beat it, so its remaining stages were not selected—this is the mechanism behind the main-text "curriculum did not help" result. Its training log shows 19 of 30 planned epochs: stage 1 ran 7, stage 2 ran 6 (absolute epochs 8–13) and stage 3 ran 6 (absolute epochs 14–19), and the best validation macro-F1 was achieved at absolute epoch 1, before any of the later stages contributed.

Second, the per-class thresholds *lowered* M4's test macro-F1: re-argmaxing the same frozen posteriors gives 0.6602, while the reported (thresholded) headline is 0.6590—the searched thresholds had raised validation macro-F1 (0.6942 $\rightarrow$ 0.6965) but did not transfer to the test split. This is why M4's "argmax M-F1" in Table 10 is higher than its headline.

Third, M5 and M6 used the full fixed 15-epoch budget with the early-stopping criterion never firing ("stopped = no": their patience is at least the budget), unlike M1–M4.

Table 11. Resolved per-system configuration as recorded in each run's `training_summary.json` and `configs/*.yaml`. "lr" is the stage-1 encoder rate; "head lr" the discriminative head rate when DLR is on. "thr." marks the systems whose test pass applied per-class thresholds searched on validation (M3–M6, read from each run's `metrics.json`); the search returned the trivial 0.5 for all but M4. AdamW, weight decay 0.01, gradient clipping 1.0 and max length 256 are shared.

System	head	stages	loss	lr	head lr	warmup	bs×acc	pat.	seed	thr.
M1	standard	1	CE	$3.0e-05$	$3.0e-05$	0.1	16×2	6	42	no
M2	custom	1	CE	$3.0e-05$	$3.0e-05$	0.1	16×2	6	42	no
M3	custom	2	CE $\rightarrow$ focal	$2.0e-05$	$1.0e-04$	0.1	16×2	6	42	yes
M4	custom	3	concept $\rightarrow$ CE	$2.0e-05$	$8.0e-05$	0.1	16×2	6	42	yes
M5	custom	1	CE	$2.0e-05$	$1.0e-04$	0.1	16×2	15	42	yes
M6	standard	1	focal	$1.0e-05$	$1.0e-05$	0.05	16×2	20	42	yes

Table 12. Wall-clock cost and checkpoint selection per system. "best ep." is the epoch whose validation macro-F1 was frozen for the test pass and "stopped" whether early stopping fired before the planned epoch budget.

System	epochs	planned	best ep.	stopped	wall clock
M1	15	30	9	yes	02:50:03
M2	20	30	14	yes	03:50:56
M3	21	30	15	yes	03:56:46
M4	19	30	1	yes	03:38:41
M5	15	15	3	no	02:48:28
M6	15	15	4	no	02:59:21

13 Complementary Experiments

Two further experiments are provided as runnable scripts but are *not* part of the main results, because ensuring they were meaningful (rather than noise) would require compute beyond a single training session.

- **Seed variance** (`scripts/09_seed_variance.py`) re-trains each of the six recipes under five seeds (30 runs) so that a seed-variance floor can be placed under the 0.654–0.664 macro-F1 band. On the available single-GPU budget this is roughly 60 h of training and was out of scope; the script is included so the experiment can be reproduced, and the main text is careful not to *claim* a seed-robust ordering.
- **Monolingual encoders** (`scripts/10_train_monolingual_baselines.py`), which trains PhoBERT (`vinai/phobert-base-v2`) and ViSoBERT (`uitnlp/visobert`) under the same recipe as the baseline, to test whether the multilingual backbone is actually doing work. This is a separate comparison and is deliberately kept out of the six-system ablation so as not to confound the optimisation-recipe study.

14 Reproduction

All tables and figures in this supplement are generated from the frozen artefacts by

```
python scripts/11_supplementary_analyses.py
```

which reads the per-system `test_predictions.csv`, `metrics.json` and `training_summary.json` under `outputs/evaluation/per_model/` and writes `papers/supplementary/tables/*.tex`, `papers/supplementary/figures/*.png` and a machine-readable summary at `outputs/evaluation/analysis/supplementary_report.md` and `supplementary_summary.json`. The frozen run snapshot `outputs/evaluation/per_model/<system>/config.yaml` is treated as authoritative for each evaluated system: the top-level `configs/*.yaml` may have been edited after a run (this is the case for M6, whose on-disk config now shows the threshold search disabled although the evaluated run enabled

it). The bootstrap uses 10,000 resamples with a fixed seed, so the intervals are deterministic given the frozen predictions. This document is then compiled with `pdflatex` + `bibtex` in `papers/supplementary/`.

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